TN-IMPACT
Tamil Nadu Innovation Modernization through Problem Solving and Advanced Creative Tech

TIDCO

DASSAULT SYSTEMES

My Problem Statement

Problem Statements

1 Problem statements

Posted On 1 month ago

Hardware/Software

Valve Assembly and Quality Control

Change Problem Statement

TNI26175- Innovation in Eliminating Assembly Errors in Multi-Component Valve Manufacturing: Description Multi-component valves involve numerous parts (seals, seats, fasteners, springs, actuators). Assembly relies largely on operators, resulting in missing components, wrong orientation, incorrect torque, and occasional safety and performance failures.

Venue- (KIT - Kalaignar Karunanidhi Institute of Technology)-Event Date: 14-03-2026

Requirements

Standard work instructions with visual guidance for each

valve model.

Poka-yoke (error-proofing) in fixtures, part bins, and fastener selection.

Torque-controlled tools with digital confirmation.

Optionally, electronic assembly checklist linked to valve serial number.

Implementation

Create or digitize step-by-step assembly procedures with pictures/3D views.

Design assembly fixtures that only accept the correct part/orientation.

Use torque wrenches or nutrunners with set limits and logging.

Introduce a simple assembly HMI/tablet that verifies each step and logs completion.

3D Models attached

1 Teams Selected

Ask Query

Total Queries : 0

Total Replies : 0

Submit Solution

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